calllib

Call function in shared library

Syntax

```
[x1, ..., xN] = calllib('libname', 'funcname', arg1, ..., argN)
```

Description

`[x1, ..., xN] = calllib('libname', 'funcname', arg1, ..., argN)` calls the function `funcname` in library `libname`, passing input arguments `arg1` through `argN`. `calllib` returns output values obtained from function `funcname` in `x1` through `XN`.

All scalar values returned by MATLAB are of type `double`.

If you used an alias when initially loading the library, then you must use that alias for the `libname` argument.

Ways to Call calllib

The following examples show ways calls to `calllib`. By using `libfunctionsview`, you determined that the `addStructByRef` function in the shared library `shrlibsample` requires a pointer to a `c_struct` data type as its argument.

Load the library:

```
addpath([matlabroot '\extern\examples\shrlib'])
loadlibrary shrlibsample shrlibsample.h
```

Create a MATLAB structure:

```
struct.p1 = 4; struct.p2 = 7.3; struct.p3 = -290;
```

Use `libstruct` to create a C structure of the proper type (`c_struct`):

```
[res,st] = calllib('shrlibsample','addStructByRef',...
        libstruct('c_struct',struct));
```

Let MATLAB convert `struct` to the proper type of C structure:

```
[res,st] = calllib('shrlibsample','addStructByRef',struct);
```

Pass an empty array to `libstruct` and assign the values from your C function:

```
[res,st] = calllib('shrlibsample','addStructByRef',...
        libstruct('c_struct',[]));
```

Let MATLAB create the proper type of structure and assign values from your C function:

```
[res,st] = calllib('shrlibsample','addStructByRef',[]);
```

Remove the library from memory:

```
unloadlibrary shrlibsample
```

Examples

To call functions in the MATLAB `libmx` library, see Invoking Library Functions.

See Also

`libfunctions` | `libfunctionsview` | `loadlibrary` | `unloadlibrary`

How To

- Passing Arguments
- MATLAB Crashes Making a Function Call to a Shared Library

© 1984-2012- The MathWorks, Inc. - Site Help - Patents - Trademarks - Privacy Policy - Preventing Piracy - RSS